```python
#Skillcraft Technology
#Traffic Accident Analysis
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

# India Population Distribution by Age Group (2022)
# Source: UN World Population Prospects 2022

age_groups = [
    '0-4', '5-9', '10-14', '15-19', '20-24',
    '25-29', '30-34', '35-39', '40-44', '45-49',
    '50-54', '55-59', '60-64', '65-69', '70-74',
    '75-79', '80+'
]

population_millions = [
    114.3, 124.7, 127.9,
    127.1, 122.4,
    118.6, 111.2, 101.5,
    90.8, 78.3,
    65.4, 54.2,
    43.8, 33.1,
    22.6, 13.2, 9.8
]

# Color categories
colors = []

for ag in age_groups:
    start = int(ag.split('-')[0].replace('+', ''))

    if start < 15:
        colors.append('#F5C518')      # Young
    elif start < 65:
        colors.append('#3A86FF')      # Working
    else:
        colors.append('#FF006E')      # Senior

fig, ax = plt.subplots(figsize=(14,7))

fig.patch.set_facecolor('#1a1a2e')
ax.set_facecolor('#16213e')

bars = ax.bar(
    age_groups,
    population_millions,
    color=colors,
    edgecolor='#0f3460',
    linewidth=0.8
)

# Labels on bars
for bar, value in zip(bars, population_millions):

    ax.text(
        bar.get_x()+bar.get_width()/2,
        value+1,
        f'{value:.1f}M',
        ha='center',
        fontsize=8,
        color='white',
        fontweight='bold'
    )

# Totals
young_total = sum(population_millions[:3])
working_total = sum(population_millions[3:13])
senior_total = sum(population_millions[13:])
total_population = sum(population_millions)

# Title
ax.set_title(
    "India Population Distribution by Age Group (2022)",
    fontsize=16,
    color='white',
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    fontweight='bold'
)

ax.set_xlabel(
    "Age Group",
    fontsize=12,
    color='white'
)

ax.set_ylabel(
    "Population (Millions)",
    fontsize=12,
    color='white'
)

ax.tick_params(colors='white')

ax.grid(
    axis='y',
    linestyle='--',
    alpha=0.4
)

# Legend
legend = [

    mpatches.Patch(
        color='#F5C518',
        label=f'Young (0-14): {young_total:.1f} M'
    ),

    mpatches.Patch(
        color='#3A86FF',
        label=f'Working (15-64): {working_total:.1f} M'
    ),

    mpatches.Patch(
        color='#FF006E',
        label=f'Senior (65+): {senior_total:.1f} M'
    )
]

ax.legend(
    handles=legend,
    facecolor='#0f3460',
    edgecolor='white',
    labelcolor='white'
)

# Info box
ax.annotate(
    f"Total Population: {total_population:.1f} Million\nMedian Age: 28 Years\nSource: UN WPP 2022",
    xy=(0.02,0.95),
    xycoords='axes fraction',
    fontsize=9,
    color='white',
    bbox=dict(
        boxstyle='round',
        facecolor='#0f3460',
        edgecolor='white'
    )
)

plt.tight_layout()

plt.savefig(
    "india_population_distribution_2022.png",
    dpi=300,
    bbox_inches='tight'
)

plt.show()

print("Chart saved successfully!")
print(f"Young: {young_total:.1f} Million")
print(f"Working: {working_total:.1f} Million")
print(f"Senior: {senior_total:.1f} Million")
print(f"Total Population: {total_population:.1f} Million")
```

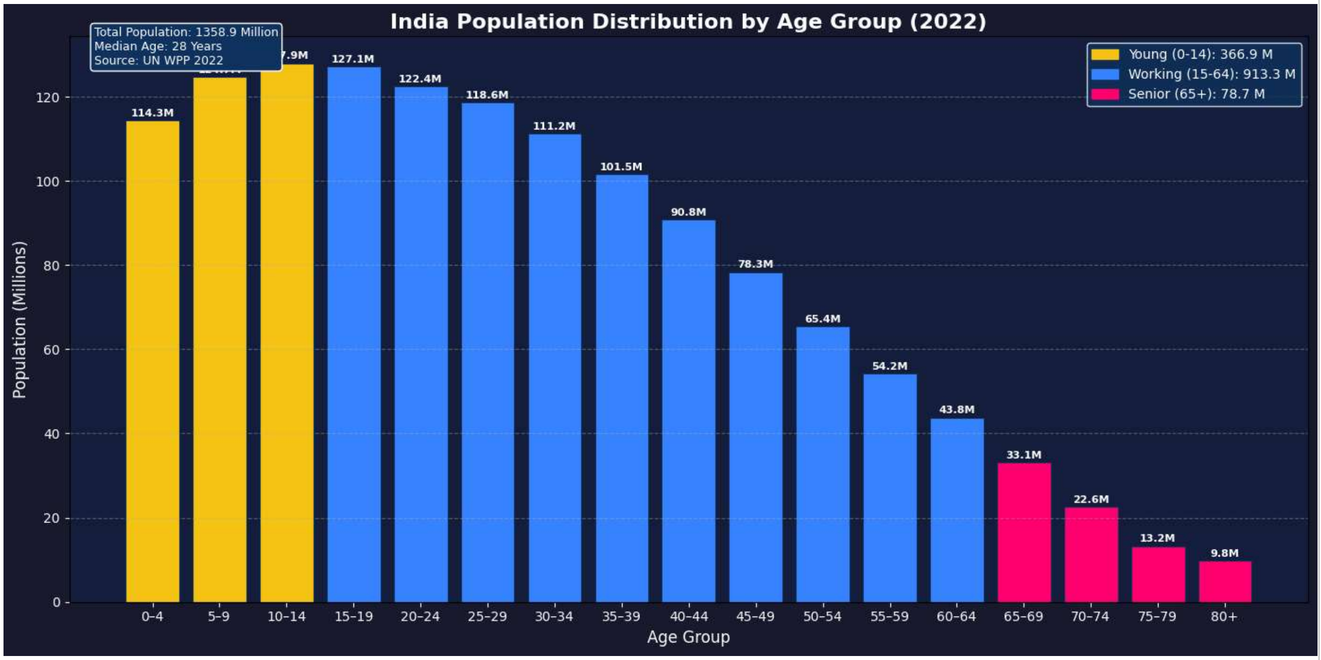


```
Chart saved successfully!
Young: 366.9 Million
Working: 913.3 Million
Senior: 78.7 Million
Total Population: 1358.9 Million
```